

Energy Resonance Coupling in Linguistics: A Perspective on the Origin of Characters through Metasurface Technology

Xinmin Zhao, Feng Zhang*

College of Foreign Languages, Terahertz Technology Innovation Research Institute, Terahertz Spectrum and Imaging Technology Cooperative Innovation Center, Shanghai Key Lab of Modern Optical System, University of Shanghai for Science and Technology, Shanghai 200093, People's Republic of China

Corresponding Email: fzhang@usst.edu.cn

Abstract

Character patterns have been regarded as mystical mediums capable of influencing the surrounding energy fields since ancient times, often used for altering environmental energy or protecting individuals. Modern metasurfaces, on the other hand, are two-dimensional structures designed with nanometer or micrometer precision to achieve highly controlled manipulation of electromagnetic waves. This paper aims to explore the connection between character patterns and metasurfaces in aspects such as energy regulation, information encoding, and cultural symbolism, while further analyzing their multidimensional impact on technology, culture, ethics, and practical applications. By comparing the historical background, cultural significance of character patterns, and the physical principles of modern metasurfaces, the paper reveals commonalities between the two in energy resonance, information transmission, and geometric symmetry. Furthermore, the paper delves into the role of character patterns in psychology and socio-culture, as well as their inspirational significance for metasurface technology design. Finally, the paper discusses the ethical and socio-cultural adaptability of metasurface technology, providing valuable insights for future scientific research and practical applications.

Keywords: Character pattern, metasurface, energy resonance, geometric symmetry, information encoding, technology ethics.

1. Introduction

Two-dimensional patterns are both symbols of human culture and tools for scientific

exploration. From the evolution of symbols into writing (hereafter referred to as characters), to the discovery of interference fringes and Fourier transforms, planar patterns have carried the fusion of information and aesthetics, reflecting the deep laws of nature. The power of planar patterns lies in their ability to connect culture with science, while also inspiring humanity's understanding of the world. Pictographs are among the earliest forms of characters, usually representing objects from the natural world in image form. These symbols are typically designed based on the shape or features of the objects they represent, making it easier for people to recognize and understand. For example, the Chinese character "日" (sun) resembles the circular shape of the sun, symbolizing light and daytime; while the character "月" (moon) looks like a crescent moon, symbolizing the moon at night. The similarity between the shapes and the objects they represent makes pictographs easy to remember and recognize. Their origin is closely related to the communication needs of early humans. Early pictographs were used not only for recording information but also frequently associated with religion and rituals. The shapes in characters were often variants of pictographs with symbolic meaning. When writing characters, the symbols are not merely combinations of letters but convey sacred intentions through specific shapes and arrangements. Therefore, characters can be seen as a graphical language with both religious and cultural significance.

"Fu" (符), generally refers to mysterious symbols made with strokes, often combining dots and lines, with curving strokes as the main feature. These symbols are believed to possess supernatural powers, such as driving away spirits, curing diseases, and resolving disasters (毕旭玲, 2006). "Zhou" (咒), on the other hand, refers to specific phrases chanted aloud, often used in conjunction with symbols, seals, and ritual instruments, producing special audio effects that can play a role in the spellcasting process (盖建民, 1999). For example, the "厥口诅祝" (verbal incantation) mentioned in the Book of Documents is a form of communication with deities to achieve specific purposes. Characters hold an important place in Taoist practices, where many sects and practitioners use them for health maintenance, blessing, and disaster prevention. This practice not only involves symbols and incantations but also complex rituals and individual cultivation (温茂兴, 2006). Moreover, the character of the pig in Chinese culture symbolizes wealth and good fortune, reflecting the desire for prosperity.

The formation of characters is closely tied to ancient religions. In early societies, for example,

the mandala, a sacred pattern originating in Hinduism and Buddhism, was commonly used in meditation and religious rituals. The form and content of characters have evolved over different historical periods and regions. As culture spread, characters gradually became accepted by societies and integrated into religious systems such as Taoism. Taoist characters, in particular, were formed through the fusion of various ancient Chinese cultural elements, including shamanism, Huang-Lao Taoism during the Warring States period, and the Way of the Immortals (Trumpf) , 1995). In ancient Egypt, characters were primarily used for protection and blessings. Hieroglyphs were widely applied in tombs and religious rituals, with common symbols like the "Ankh" (key of life) and "Uraeus" (protective symbol), which symbolized life, resurrection, and protection. The use of characters in pharaoh's tombs was particularly important to ensure the safety of the deceased in the afterlife. In India, characters were seen as tools for communicating with deities, such as the "Om" symbol, representing the sound and creativity of the universe. In the West, especially in medieval Europe, characters were closely associated with magic. Many symbols were believed to bring good luck, love, or wealth, with common symbols such as the four-leaf clover and horseshoe.

Character patterns embody the diversity across cultural backgrounds and the symbolic significance of energy control, carrying profound beliefs and psychological projections. From the hieroglyphs of ancient Egypt to the Taoist characters of China, these patterns reflect humanity's reverence for natural forces and the exploration of the unknown. Characters are not only a part of religion and ritual but also an important medium in social and personal life, often used as amulets, to convey blessings, or to dispel negative energy. Character patterns are seen as having protective, healing, and energy-guiding functions, often used in prayers for good luck, exorcism, disease prevention, and spiritual protection. Throughout history, characters have been closely linked to religious beliefs, witchcraft, and traditional medicine, gradually evolving into a symbolic system that carries rich historical and social significance. As characters spread and evolved in different cultures, their forms and functions have changed, showcasing a common human desire to control fate and obtain security. Studying these symbols not only helps us understand mystical traditions but also reveals their redefinition and application in modern society.

Not only do character patterns rely on two-dimensional designs, but writing—whether pictographs or Arabic numerals—can also be seen as a type of planar pattern. Discoveries in physics,

such as interference fringes (Shi et al., 2014; Wang et al., 2020a), diffraction patterns (Du et al., 2005; Yi et al., 2014), and related Fourier transform algorithms (Braiman and Rothschild, 1988), highlight the far-reaching significance of planar patterns in human cognition and exploration. In recent years, metasurface technology has emerged as another important innovation related to two-dimensional patterns in physics and engineering (Guddala et al., 2021; Hu et al., 2019; Li et al., 2022). Metasurface technology achieves precise control over energy forms such as light and electromagnetic waves through the fine design of nanostructures. Metasurfaces, which have the characteristic of two-dimensional planar structures, have been widely applied in optical information processing, sensor design, stealth technology, and energy management in recent years (Zang et al., 2021). The basic principles of metasurface technology include precise phase control of electromagnetic waves and optimization of energy absorption and reflection, which offer unique advantages in the manufacture of miniature antennas, optical filters, and stealth materials. By designing the microstructure of materials, metasurfaces enable customized manipulation of electromagnetic waves on the nanoscale, overcoming the performance limitations of traditional optical elements. Furthermore, their application in energy management, such as thermal radiation control and wireless energy transfer, provides vast potential for the development of smart devices and responsive environmental systems. Additionally, the development of metasurfaces has driven advancements in frontier fields such as optoelectronic integration and quantum information processing, showcasing its core position in modern technology (Ma et al., 2022).

Character patterns and metasurfaces exhibit similarities in energy control. Characters, through specific geometric forms, induce resonance phenomena that enhance the perceiver's awareness of energy changes, while metasurfaces achieve resonance and modulation of electromagnetic waves at specific frequencies through nanostructural design. This resonance phenomenon reveals common characteristics in their structures' influence on energy distribution, indicating a deep connection between symbols and physical designs in terms of energy concentration and scattering. Some of these phenomena have been validated through our research on the resonance coupling between optical microcavities and biological responses (Gao et al., 2023; Gu et al., 2023; Hou et al., 2024; Li and Zhang, 2023; Niu et al., 2024; Si et al., 2024; Wang et al., 2024; Zhang et al., 2023; Zhong et al., 2023). Geometric symmetry is also a key link between characters and metasurfaces. The

symmetrical design in character patterns aims to create psychological and spiritual stability, while the symmetry in metasurfaces is used to precisely control the propagation of electromagnetic waves, enabling energy focusing or scattering. The structural similarities between the two provide a new perspective on how energy concentration and guidance can be achieved through specific designs. Additionally, information encoding (Chang et al., 2004) is another commonality between characters and metasurfaces. Characters convey specific cultural information through patterns, while metasurfaces modulate information by adjusting optical phases and polarization states (Chen et al., 2022). This ability to convey information through symbols not only reflects the complexity of the symbols but also provides inspiration for modern technology in the fields of information encryption and communication.

The combination of character patterns and metasurfaces not only theoretically reveals the physical potential of characters but also offers new directions for the development of modern science and technology. Exploring the commonalities and intersections between characters and metasurfaces helps scientifically explain the energy effects of characters and holds the potential to apply the wisdom of ancient symbolism to modern technological design, promoting a deep integration of science and culture.

2. The Relationship Between Character Patterns and Metasurfaces

Character patterns and metasurfaces exhibit similarities in the concept of "resonance," as both influence energy distribution through geometric or structural design. Character patterns induce potential resonance phenomena through their shapes, enhancing the perceiver's awareness of energy changes, while metasurfaces achieve resonance and modulation of electromagnetic waves at specific frequencies through the design of nanostructures (Aigner et al., 2022). The physical basis of this resonance behavior lies in how structural dimensions and shapes impact energy fluctuations, similar to the intricate geometric arrangements of lines in character patterns, which produce energy concentration or scattering effects through specific designs. This energy resonance phenomenon's similarity may reveal the potential physical principles behind character patterns, providing a scientific explanation for the "power" of ancient symbols. Resonance plays a key role in both characters and metasurfaces. For example, certain character patterns are believed to guide or amplify the human bioenergy field, a concept that can be likened to electromagnetic field energy

accumulation in resonance cavities (Liu et al., 2022; Malak et al., 2013; Zhao et al., 2024). By designing the geometric shapes of characters, it is possible to achieve specific frequency responses, much like how metasurfaces tune electromagnetic waves at particular frequencies using geometric parameters.

Character patterns and metasurfaces focus on different aspects of geometric symmetry in design, but both exhibit common traits in guiding energy concentration and stability. Character patterns, through meticulously designed symmetrical shapes, aim to evoke psychological and spiritual effects. Their symmetry is often closely linked to concepts of natural and cosmic order. For example, the Taiji diagram in China and mandalas both use complex geometric symmetry to symbolize the balance and harmony of the universe (Table-1). Similarly, the symmetry in metasurface designs affects the scattering and absorption characteristics of electromagnetic waves, and through careful symmetrical structure design, precise control of energy focusing, absorption, and reflection can be achieved (Yao et al., 2021; Zhang et al., 2021). By analyzing the physical effects of the geometric symmetry in both, one can further understand how energy concentration, deflection, and guidance can be achieved through specific patterns to accomplish particular functions. Comparing the geometric designs of characters and metasurfaces can provide inspiration for modern scientific design, combining tradition with technology to explore more effective energy control solutions.

Geometric symmetry is at the core of both character patterns and metasurface design. For example, the multilayered symmetry of the mandala pattern not only brings a sense of tranquility and harmony on a spiritual level but also reveals a profound aesthetic of symmetry and balance. This can guide individuals into a meditative and reflective state (毗卢遮那佛（大日如来），唐朝开元年间（公元 724-725 年）). At the same time, this complex symmetrical structure may have significant implications for energy transfer. Through its geometric arrangement and the visual guidance of energy flow, it helps to form a positive resonance effect between the viewer and the surrounding environment, enhancing focus, reducing anxiety, and fostering inner balance and integration. Clearly, these patterns affect the mental state, possibly through subtle actions on the nervous system. Recent studies have found that electromagnetic waves, especially terahertz lasers, can significantly affect neuronal activity and animal behavior (Liu et al., 2021; Tan et al., 2023;

Wang et al., 2020b; Xiao et al., 2023). The symmetry design in metasurfaces similarly influences the paths of photons or electrons, determining the efficiency and stability of energy transmission. This design's potential applications in energy conduction, conversion, and storage, combined with the geometric symmetry of traditional symbolism, can offer new perspectives for energy management technologies.

Character patterns convey cultural signals through graphical language, while metasurfaces can achieve information encoding through optical phases, polarization, and other methods. The metaphors and semiotic information embedded in characters often imply the presence of protection, blessings, or power, carrying rich socio-cultural and psychological significance. Modern metasurfaces modulate information by controlling the phases of electromagnetic waves, offering highly precise encoding capabilities. For example, metasurfaces can encode the polarization state of light for encrypted communication and information steganography. From one perspective, both characters and metasurfaces possess the ability to convey abstract information through specific symbols, with the medium shifting from ancient patterns to modern nanostructures. Further combining the concept of characters with metasurface technology could open up unique quantum encryption techniques based on semiotics, leveraging the diversity of ancient symbols to increase the complexity and security of information encoding. Information encoding is a key element in both character patterns and metasurface technologies. For instance, complex lines and symbols in character patterns are used to express abstract concepts and energies, which have profound spiritual meaning for the believers. Modern science encodes light's phases and polarization in metasurfaces to transmit complex information. By combining characters with metasurfaces, one can use the unique visual features of symbols to encrypt information, especially in applications requiring cultural sensitivity (e.g., cultural heritage preservation, historical archive management), enabling more culturally meaningful information transmission and protection.

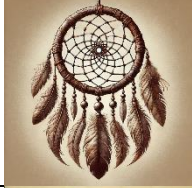
3. Deepening the Cultural Significance of Character Patterns

Character patterns have unique forms of expression globally. For example, in Eastern cultures, symbols such as the Chinese Bagua diagram, the Five Elements, and the Japanese mandala structure are widely present, symbolizing the balance and flow of energy. In Western cultures, Celtic knots, ancient runes, and crosses represent protection and strength. These symbols guide the flow of energy

through their distinctive geometric shapes, presenting energy resonance and transformation through different design languages. Analyzing the specific symbolism and practical functions of these symbols in their cultural context helps to understand how they establish a connection between humans and energy through visual representation, forming a symbolic "energy field." The diversity and symbolism of character patterns also provide insights for the cultural adaptability of modern technology in global promotion. Combining symbolic designs from different cultures can enhance the acceptance and popularity of technology. Behind character patterns in different cultures often lies profound philosophy and worldviews. For instance, the Chinese Bagua diagram is not just a character but also a description of the balance of yin and yang in the natural world, symbolizing mutual generation and restraint. Similarly, the Celtic knot, through its endless loop, symbolizes the eternity of life and the protection of the spirit. Delving deeper into the cultural significance of these symbols can not only deepen the philosophical understanding behind them but also explore their potential application value in modern technology design.

Table 1: Comparison of Spiritual Patterns in Different Religious Cultures.

Religion/Culture	Pattern Name	Usage	Spiritual Effect	Pattern Characteristics	Typical Pattern
Tibetan Buddhism	Mandala	Used for meditation and visual contemplation to achieve inner peace	Helps to enter a meditative state, enhances inner calm and spirituality	Multilayered symmetrical geometric and symbolic design, rich in colors	
Hinduism	Yantra	Used for meditation and worship, focuses the mind	Focuses the mind, connects with universal energy	Geometric patterns combining triangles, circles, and squares	
Islamic	Ghirih	Used in architectural decoration and ornaments, aids concentration during meditation	Symbolizes the sacred and the infinite, helps with focus	Symmetrical geometric patterns including polygons and stars	
Christianity	Rose Window	Used in churches to guide meditation and reflection	Symbolizes divine light and cosmic harmony, aids in meditation	Radiating, colorful geometric patterns with stained glass windows	

Native American	Dreamcatcher	Hung in sleeping areas to protect from nightmares	Brings peaceful sleep and protection	Circular structure with a webbed center, symbolizes cycles and protection	
Celtic Culture	Celtic Knot	Used as decorative symbols, representing eternity and spiritual connection	Symbolizes eternity, spirituality, and the cycle of life, provides a sense of tranquility	Continuous woven ring structure with no clear start or end	
Daoism	Bagua	Used in Feng Shui, philosophical study, and meditation guidance	Symbolizes the harmony of Yin and Yang, helps with focus and spiritual balance	Central Yin-Yang symbol, surrounded by eight trigrams representing natural elements	

Character patterns are not only perceived as possessing power on a physical level, but their psychological effects are also worthy of attention. Research shows that trust in character patterns can trigger the placebo effect, leading to psychological comfort and improved states. This effect demonstrates the profound influence that character patterns have on the cultural and psychological level, suggesting an intrinsic connection between symbols, belief, and energy. From the perspective of neuroscience, these character patterns may influence the human endocrine system by enhancing trust and belief, thereby generating positive physiological responses. A deeper exploration of the psychological mechanisms behind character patterns can further reveal the important role that cultural beliefs play in physical health and psychological stability. By integrating the psychological effects of character patterns with modern psychological therapies, such as cognitive behavioral therapy (CBT), new therapeutic approaches can be introduced into the mental health field, using symbols to enhance the power of suggestion and improve treatment efficacy. The psychological effects of character patterns can also be applied to stress management and anxiety relief. In specific cultural contexts, character patterns are considered to have a calming function for the mind. For example, the creation of Mandalas in Tibetan Buddhism is a meditative and therapeutic process that helps individuals achieve inner peace by focusing on the creation and observation of complex patterns. By combining modern psychological therapy techniques, a set of symbol-based therapeutic interventions can be designed to treat psychological health issues such as anxiety and depression, enhancing therapeutic outcomes through symbolic guidance and energy perception.

4. Exploring Ethical and Social Impacts

As the application of metasurfaces expands in fields such as sensors and information processing, ensuring their safe and ethical use has become a significant concern. For example, metasurfaces can be used in advanced biological monitoring systems (Chen et al., 2022; Xu et al., 2021; Xu et al., 2023), but without proper regulation, they may infringe on privacy. When designing and applying metasurface technologies, ethical considerations and their potential societal impact must be carefully addressed to prevent misuse. Moreover, the development of these technologies requires establishing relevant standards and legal frameworks to safeguard public privacy and data security. Transparency and controllability should be core principles in data collection and processing to ensure that technological progress does not negatively affect society. At the same time, it is crucial to ensure that any collection and use of personal data in metasurface applications is done with explicit consent, to prevent potential misuse. Ethical considerations also include the fairness and accessibility of the technology. If metasurface technology is used in critical fields such as healthcare and communications, it could exacerbate technological inequality, particularly in underdeveloped regions. Therefore, when developing and promoting metasurface technology, policies should be implemented to ensure its inclusivity and fairness, enabling more people to benefit from technological advancements.

In the context of character patterns, applying metasurface technology in culturally sensitive traditional areas requires respect for different cultural contexts. For instance, when combining metasurface design with traditional character patterns in cultural sites or memorial objects, it is important to understand and respect local cultural beliefs to ensure that technology and culture evolve harmoniously. Analyzing the adaptability of different cultural symbols can help avoid cultural conflicts in technological applications and enhance the global acceptance of the technology. Cultural differences affect the public's receptiveness to new technologies, so when promoting metasurface technology, these differences should be considered, and appropriate communication strategies should be developed to gain support and understanding from various regions. For example, in religious or historically symbolic locations, collaboration with local cultural experts is essential to ensure that the use of technology aligns with cultural traditions. In combining character patterns with metasurface technology, cultural adaptability is particularly important. In some cultural

contexts, characters may be viewed as having strong sacred significance, and their integration with modern technology could provoke resistance from believers. Therefore, in practical applications, cultural consultation and community involvement should ensure the legitimacy and societal acceptance of the technology, thereby facilitating the fusion of science and culture.

5. Conclusion

This paper explores the deep connections between character patterns and modern metasurface technologies, particularly the similarities in energy control, information encoding, and geometric design. Through in-depth analysis of the cultural connotations of character patterns, their psychological effects, and experimental verification and application cases of metasurface technology, the paper reveals how character patterns inspire modern technological designs and suggests the possibility of integrating characters with metasurfaces to achieve a fusion of culture and technology. Additionally, the paper delves into the ethical considerations and socio-cultural adaptability of metasurface technology, emphasizing the importance of recognizing cultural diversity and societal acceptance during technology promotion. Future research can further investigate how to validate the potential physical effects of character patterns within modern scientific frameworks and explore better ways to integrate them with metasurface technology, thus advancing the development of new cultural semiotics technologies. This interdisciplinary research not only helps uncover the scientific principles behind ancient symbols but also provides new ideas and inspirations for modern technological applications, particularly in areas such as cultural preservation, healthcare, and communication security. By closely combining science and culture, we can create technologies with greater social significance and cultural value, driving the fusion of science and the humanities.

References

- Aigner, A., Tittl, A., Wang, J., Weber, T., Kivshar, Y., Maier, S.A., Ren, H., 2022. Plasmonic bound states in the continuum to tailor light-matter coupling. *Sci Adv* 8, eadd4816.
- Braiman, M.S., Rothschild, K.J., 1988. Fourier transform infrared techniques for probing membrane protein structure. *Annual review of biophysics and biophysical chemistry* 17, 541-570.

Chang, S., Zhou, M., Grover, C.P., 2004. Information coding and retrieving using fluorescent semiconductor nanocrystals for object identification. *Optics express* 12, 143-148.

Chen, B., Wang, X., Li, W., Li, C., Wang, Z., Guo, H., Wu, J., Fan, K., Zhang, C., He, Y., Jin, B., Chen, J., Wu, P., 2022. Electrically addressable integrated intelligent terahertz metasurface. *Sci Adv* 8, eadd1296.

Du, C., Falini, G., Fermani, S., Abbott, C., Moradian-Oldak, J., 2005. Supramolecular assembly of amelogenin nanospheres into birefringent microribbons. *Science* 307, 1450-1454.

Gao, F., Guo, J., Si, Q., Wang, L., Zhang, F., Yang, F., 2023. Modification of ATP hydrolysis by Strong Coupling with O–H Stretching Vibration. *ChemPhotoChem* 7, e202200330.

Gu, K., Si, Q., Li, N., Gao, F., Wang, L., Zhang, F., 2023. Regulation of Recombinase Polymerase Amplification by Vibrational Strong Coupling of Water. *ACS photonics* 10, 1633-1637.

Guddala, S., Komissarenko, F., Kiriushchikina, S., Vakulenko, A., Li, M., Menon, V.M., Alu, A., Khanikaev, A.B., 2021. Topological phonon-polariton funneling in midinfrared metasurfaces. *Science* 374, 225-+.

Hou, S., Gao, F., Zhong, C., Li, J., Zhu, Z., Wang, L., Zhao, Z., Zhang, F., 2024. Vibrational Alchemy of DNA: Exploring the Mysteries of Hybridization under Cooperative Strong Coupling with Water. *ACS photonics* 11, 1303-1310.

Hu, G., Hong, X., Wang, K., Wu, J., Xu, H.-X., Zhao, W., Liu, W., Zhang, S., Garcia-Vidal, F., Wang, B., Lu, P., Qiu, C.-W., 2019. Coherent steering of nonlinear chiral valley photons with a synthetic Au–WS₂ metasurface. *Nature Photonics* 13, 467-472.

Li, N., Zhang, F., 2023. THz-PCR Based on Resonant Coupling between Middle Infrared and DNA Carbonyl Vibrations. *ACS applied materials & interfaces* 15, 8224-8231.

Li, Q., van de Groep, J., White, A.K., Song, J.H., Longwell, S.A., Fordyce, P.M., Quake, S.R., Kik, P.G., Brongersma, M.L., 2022. Metasurface optofluidics for dynamic control of light fields. *Nat Nanotechnol* 17, 1097-1103.

Liu, K., Huang, G., Li, X., Zhu, G., Du, W., Wang, T., 2022. Vibrational Strong Coupling between Surface Phonon Polaritons and Organic Molecules via Single Quartz Micropillars. *Adv Mater* 34, e2109088.

Liu, X., Qiao, Z., Chai, Y., Zhu, Z., Wu, K., Ji, W., Li, D., Xiao, Y., Mao, L., Chang, C., Wen, Q., Song, B., Shu, Y., 2021. Nonthermal and reversible control of neuronal signaling and behavior by midinfrared stimulation. *Proc Natl Acad Sci U S A* 118.

Ma, Z., Li, P., Chen, S., Wu, X., 2022. Optical generation of strong-field terahertz radiation and its application in nonlinear terahertz metasurfaces. *Nanophotonics* 11, 1847-1862.

Malak, M., Gaber, N., Marty, F., Pavy, N., Richalot, E., Bourouina, T., 2013. Analysis of Fabry-Perot optical micro-cavities based on coating-free all-silicon cylindrical Bragg reflectors. *Optics express* 21, 2378-2392.

Niu, X., Wu, Z., Gao, F., Hou, S., Liu, S., Zhao, X., Wang, L., Guo, J., Zhang, F., 2024. Resonating with Cellular Pathways: Transcriptome Insights into Nonthermal Bioeffects of Middle Infrared Light Stimulation and Vibrational Strong Coupling on Cell Proliferation and Migration. *Research* 7.

Shi, L., Hakala, T.K., Rekola, H.T., Martikainen, J.P., Moerland, R.J., Torma, P., 2014. Spatial coherence properties of organic molecules coupled to plasmonic surface lattice resonances in the weak and strong coupling regimes. *Physical review letters* 112, 153002.

Si, Q., Guo, J., Lian, J., Liu, A., Zhao, X., Liu, S., Peng, R., Xu, R., Zhang, F., 2024. Multimodal competition shapes enzymatic ATP hydrolysis: Deciphering microscale confinement by vibrational strong coupling. *Chemical Engineering Journal* 496.

Tan, X., Gao, M., Chang, C., 2023. A new means of energy supply driven by terahertz photons recovers related neural activity. *Iscience* 26, 105979.

Trumpf) , 澳.加.特.G., 1995. 宗教起源探索. 四川人民出版社.

Wang, H., Janzen, E., Wang, L., Edgar, J.H., Xu, X.G., 2020a. Probing Mid-Infrared Phonon Polaritons in the Aqueous Phase. *Nano Lett* 20, 3986-3991.

Wang, L.P., Gao, F., Zhao, Z.X., Liu, S.h., Xu, R.S., Guo, J., Zhang, F., 2024. Spatiotemporal omics of life energy: Towards medicine of frequencies and terahertz drugs. *MedComm – Biomaterials and Applications* 3.

Wang, Y., Wang, G., Xu, D., Jiang, B., Ge, M., Wu, L., Yang, C., Mu, N., Wang, S., Chang, C., Chen, T., Feng, H., Yao, J., 2020b. Terahertz spectroscopic diagnosis of early blast-induced traumatic brain injury in rats. *Biomed Opt Express* 11, 4085-4098.

Xiao, T., Wu, K., Wang, P., Ding, Y., Yang, X., Chang, C., Yang, Y., 2023. Sensory input-dependent gain modulation of the optokinetic nystagmus by mid-infrared stimulation in pigeons. *Elife* 12.

Xu, J., Liao, D., Gupta, M., Zhu, Y., Zhuang, S., Singh, R., Chen, L., 2021. Terahertz Microfluidic Sensing with Dual-Torus Toroidal Metasurfaces. *Advanced Optical Materials* 9.

Xu, W., Fang, W., Shi, T., Ming, X., Wang, Y., Xie, L., Peng, L., Chen, H.-T., Ying, Y., 2023. Plasmonic Terahertz Devices and Sensors Based on Carbon Electronics. *Acs Applied Materials & Interfaces*.

Yao, B., Zang, X., Zhu, Y., Yu, D., Xie, J., Chen, L., Han, S., Zhu, Y., Zhuang, S., 2021. Spin-decoupled metalens with intensity-tunable multiple focal points. *Photonics Research* 9, 1019-1032.

Yi, J.M., Cuche, A., Devaux, E., Genet, C., Ebbesen, T.W., 2014. Beaming Visible Light with a Plasmonic Aperture Antenna. *ACS photonics* 1, 365-370.

Zang, X., Yao, B., Chen, L., Xie, J., Guo, X., Balakin, A.V., Shkurinov, A.P., Zhuang, S., 2021. Metasurfaces for manipulating terahertz waves. *Light: Advanced Manufacturing* 2.

Zhang, X., Guo, J., Zhang, F., 2023. Mid-Infrared Photons Enhance Polymerase Chain Reaction Efficiency by Strong Coupling with Vibrational DNA Molecules. *ACS photonics* 10, 751-756.

Zhang, Y., Chen, X., Cai, B., Luan, H., Zhang, Q., Gu, M., 2021. Photonics Empowered Passive Radiative Cooling. *Advanced Photonics Research* 2.

Zhao, X., Peng, H., Hu, J., Wang, L., Zhang, F., 2024. Nanotechnology-Enabled PCR with

Tunable Energy Dynamics. *JACS* 124, 3370-3382.

Zhong, C., Hou, S., Zhao, X., Bai, J., Wang, Z., Gao, F., Guo, J., Zhang, F., 2023. Driving DNA Origami Coassembling by Vibrational Strong Coupling in the Dark. *ACS photonics* 10, 1618-1623.

毕旭玲, 2006. 对语言力量的崇拜——读《道教符咒选讲》. *世界宗教文化*, 2.

盖建民, 1999. 道教符咒治病术的理性批判. *世界宗教研究*, 11.

毗卢遮那佛（大日如来），唐朝开元年间（公元 724-725 年）. 大毗卢遮那成佛神变加持经.

温茂兴, 2006. 论道教"祝由符咒"的实用价值及其对中医"意疗"的影响. *南京中医药大学学报：社会科学版* 7, 3.